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Design Review 1 Report – Battery Heat Generation Modeling

Battery Box Project Team

Sponsor: *Texas A&M Solar Car Racing Team*

Course Information: *MEEN 401 – 916*

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Glossary Table

Term	Definition
ASC	The American Solar Challenge, a biennial collegiate competition with solar-powered vehicles.
BMS	Battery management system; monitors the battery conditions to keep the cells within operating limits.
FSGP	Formula Sun Grand Prix, an annual solar-power vehicle track race
PCB	Printed circuit board; supports networks of electrical connections and components.
SNPS	Solution-neutral problem statement; describes the problem statement in accordance with customer needs without setting solutions.
SoC	State-of-Charge; the remaining charge of a battery cell at an instant in time.
TR	Thermal Runaway; the rapid and uncontrolled increases in chemical reactions within a battery cell that propagates.
PCM	Phase change material; absorbs latent heat when changing phases.

Introduction

The American Solar Challenge (ASC) is a biennial competition that is centered around collegiate teams designing and building solar cars that travel as far as possible purely using solar energy. The solar panels on a car absorb solar radiation and turn it into useful energy in the battery cells they are connected to. The battery cells then discharge to all electronic components in need, such as the DC motors that drive the car wheels, a cooling system for the driver of the vehicle, and relevant sensors. Thus, our project – the battery box – ensures that these battery cells and other electronic components are stored safely, ventilated, and cooled properly during the race on the Texas A&M Solar Car to compete in accordance with ASC regulations for our sponsor, the Texas A&M Solar Car Racing Team. At an industry level, battery boxes are essential in ensuring battery cells in electric vehicles do not overheat and are properly secured, thus enabling more carbon-neutral energy processes.

The problem is that our sponsor needs a battery box that is lightweight, filters air intake to reduce particulate matter and cools the inner components efficiently. I, the author, am reviewing existing heat-generation estimation methods, Aidan Spira is researching air filtration systems, Matthew Bower is researching air intake and exhaust designs, Edward Li is both researching battery cooling mechanisms and acting as our main point of contact for the sponsor, Quinn is researching contactors and electronics heat generation, and Langan is researching structural & crash protection. In the following report, findings related to battery heat generation modeling and its theory are discussed. It is found that as opposed to general electronics components with respective internal resistances, the heat in battery cells due to chemical and other internal processes should be accounted for. A satisfactory modeling program that has been used by multiple research teams with the ability to set boundary and initial conditions for battery cells is COMSOL, which is likely what our team will use. Likely issues we must account for in our product include thermal runaway and a decrease in our battery pack due to ageing, which the state-of-charge helps identify.

Background

There are different types of battery cells, such as alkaline, lithium-ion, and lead-acid batteries, to name a few. Each of these has its own advantages and disadvantages, with selection depending on the application [1]. For example, alkaline batteries are some of the most used kinds of non-rechargeable batteries, with sizes typically seen in house-hold items – AA, AAA, 9-Volt, etc. These have a high energy output relative to their size. Lithium-ion batteries are also a type of rechargeable battery that have significantly higher energy density than alkaline batteries. These kinds of batteries allow for high voltage operations and are good with high-drain devices. Lead-acid batteries are commonly used in automotive applications and backup power systems. However, these have a shorter lifespan and are generally very bulky.

For these reasons, our sponsor has determined that we are to work with cylindrical lithium-ion batteries, specifically, the Molicel INR-21700-M65A cells [2]. The reason for the cylindrical shape of these cells is to allow the cathode and anode electrode layers, between which lithium salts – "electrolytes" – facilitate the movement ion movement [3]. This shape allows these layers to stack as compactly as possible. With these battery cells having consistent cylindrical geometry, deriving equations for heat generation becomes easier. For an individual cell, we may use the derivation based on cylindrical coordinates as a boundary and initial value problem.

For a single battery cell, we start with the general heat-equation in cylindrical coordinates for a point on the cell.

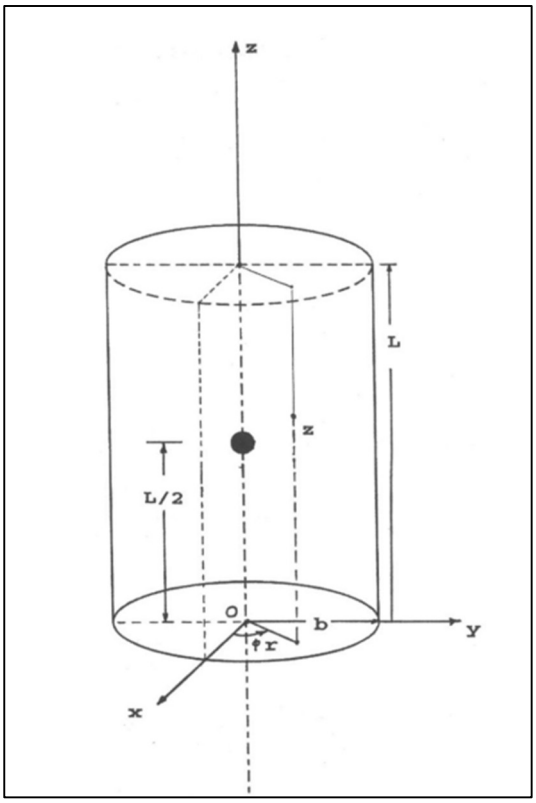


Figure 1: Image of Battery Cell represented as a Cylinder

Assuming the material is orthotropic in that thermal conductivities are different in the three orthogonal directions:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(rk_r\frac{\partial T}{\partial r}\right)+\frac{1}{r}\frac{\partial}{\partial \varphi}\left(\frac{k_\varphi}{r}\frac{\partial T}{\partial \varphi}\right)+\frac{1}{r}\frac{\partial}{\partial z}\left(rk_z\frac{\partial T}{\partial z}\right)+g(r,\varphi,z,t)=\rho c_p\frac{\partial T}{\partial t},$$

Where $g\ (\dots)$ is the battery heat generation function, ρ is the density, and c_p is the specific heat of the material. Ultimately, one of the most important variables to track for accurate heat generation is temperature, T [4]. Changes in velocity affect how temperature varies. However, it is worth noting that vehicle speed and airflow in the enclosure will be operating conditions to look further into, along with sustained driving and low-voltage operations.

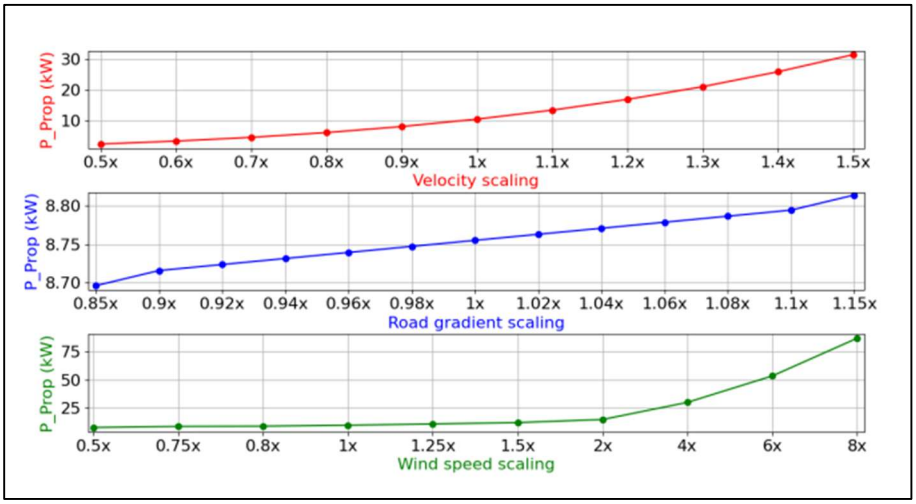


Figure 2: Plots of Power Consumption (kW) vs. Scaling of different Factors for an Electric Vehicle

Figure 2 shows graphs of an electric vehicle's power consumption vs. velocity, road gradient, and wind speed scaling. In Ganta's research, the objective was to analyze how driving and environmental parameters affect an electric vehicle's energy consumption [5]. They found the velocity to have the greatest influence on energy consumption and state-of-charge (SoC), which denotes the capacity left in battery cells at a given time and state of aging. Therefore, when the average or final velocity is higher, greater power consumption is required due to the increasing aerodynamic load, giving us a better idea of factors that increase battery heat generation. We will compare current, duration of battery, SoC, and temperature across these conditions to correctly determine heat generation.

Although we may model singular battery cells, these are always connected in groups, both in series and parallel. Rezgui's research addresses this issue by modeling groups of battery cells, as they are typically stored to save space in a typical battery box [6].

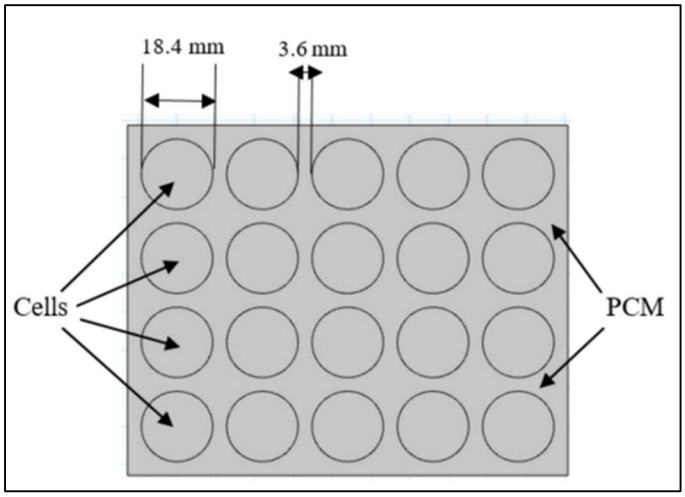


Figure 3: Representation of a Pack of Battery Cells

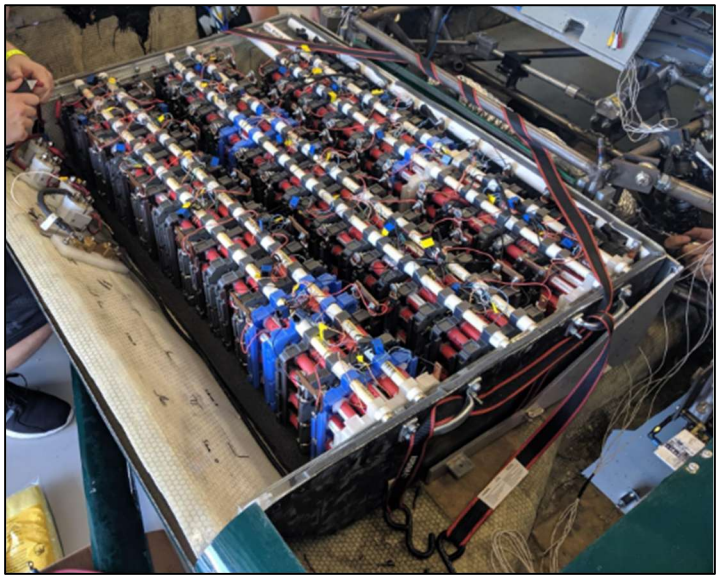


Figure 4: Michigan State Battery Cells, FSGP 2019

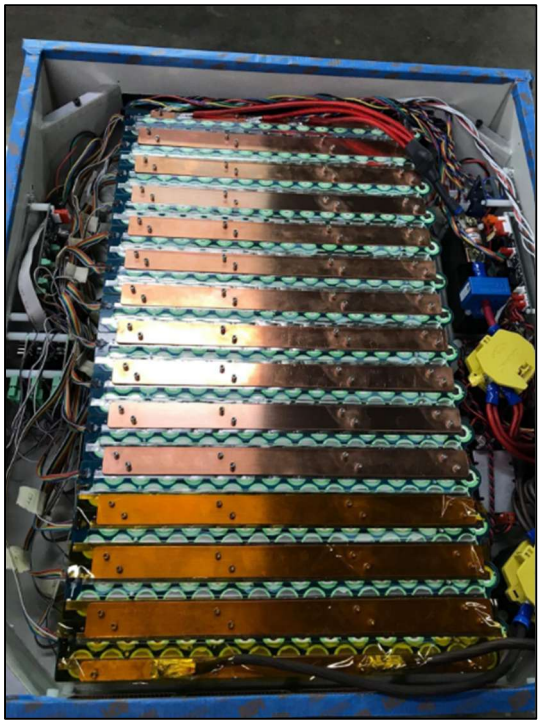


Figure 5: University of Illinois Battery Cells

Figures 4 and **5** show previous Solar Car teams' battery cells setups in the Formula Sun Grand Prix (FSGP) competitions. Multiple cells are necessary to ensure the proper wattage is achieved for each component on the car that needs energy. As is noted in these images, we are able to orient the battery modules that make up the entire battery pack in different ways to potentially accommodate our preferred method of cooling

Figure 3 displays the setup for both computer modeling and physical experiments Rezgui's team used. The objective was to develop an accurate thermal model for lithium-ion batteries that incorporated all heat sources. The setup consisted of a 20-cell battery pack under discharge rates of 1C, 3C, and 5C over a time-span of 1500 seconds. For the two-dimensional model, they account for ohmic heating, polarization losses, reversible entropy change, etc. – particularly important was accounting for the chemical side reactions, which distinguishes battery cells from other normal electronic components. These internal characteristics are not easily tracked by Battery Management Systems (BMS), used to track the voltage, current, and help estimate a SoC [7]. These modeling parameters were set in the physics engine they used – COMSOL Multiphysics 6.1, which gives us an idea for what software we could use for our modeling.

In both their experiments and modeling, they use different phase change materials (PCM)-based cooling to see how each of their temperatures change with different levels of discharge.

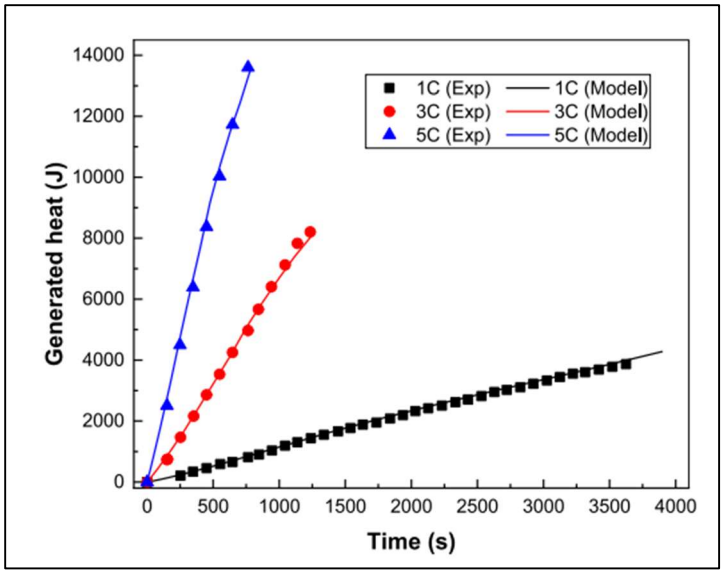


Figure 6: Experimental & Modeled Results for Generated Heat (J) vs. Time (s)

Figure 6 shows the difference between experimental and modeled results for generated heat versus time without any PCMs under different discharge conditions. This shows how both measured and experimental results had a high level of agreement.

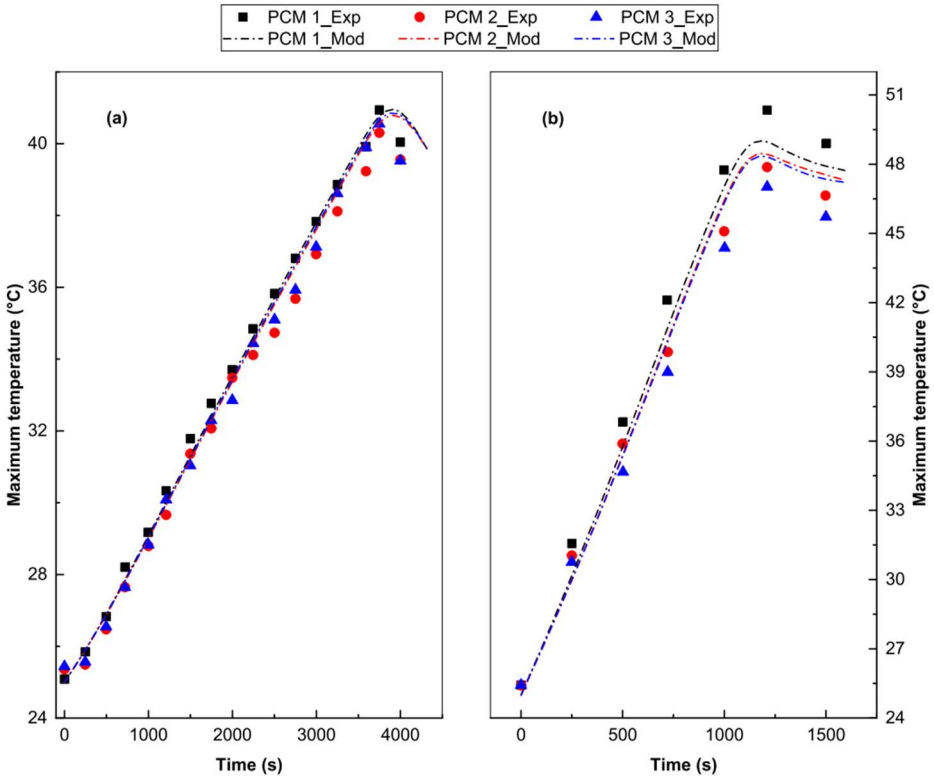


Figure 7: Plots of Maximum Temperature (°C) vs. Time (s) for 1C (left) and 3C (right) of Discharge Rates

Figure 7 also shows the high level of agreement between the measured and experimental results, where the plot on the left applies to 1C discharge conditions and the plot on the right to 3C discharge conditions, each with different PCMs. These figures prove that a mathematical model can predict with great confidence battery cell conditions, which would allow controls on the battery cells to take preemptive measures by, for example, increasing cooling, limiting current flow to the batteries, etc. The total heat generation equation their research team used included all potential sources of heat in the batteries themselves:

$$Q_{gen} = Q_{Joule} + Q_{overtop} + Q_{rev} + Q_{side} + Q_{aging} + Q_{stop}$$

With variables described in the following chart:

Mathematical expressions for the six heat generation mechanisms included in the proposed model.

Term	Description	Simplified expression	Full expression
Joule effect (Q_{Joule})	Ohmic losses due to the cell's internal resistance. Modeled as dynamic function of temperature, SOC, and current rate.	$\frac{I^2}{V_{cell}} R_{int}$	Eq. (S1.1), Supp. Mat.
Overpotential losses ($Q_{overpot}$)	Irreversible heat from polarization losses at the electrode/ electrolyte interfaces.	$\frac{I}{V_{cell}} \eta$	Eq. (S1.2), Supp. Mat.
Reversible entropic heating (Q_{rev})	Reversible heat generated by the entropy change of the intercalation/ deintercalation reactions.	$\frac{IT}{V_{cell}} \frac{dU}{dT} dU_{boost}$	Eq. (S1.3), Supp. Mat.
Side reactions' heat (Q_{side})	Exothermic heat from parasitic reactions (e.g., SEI growth, gas evolution). Modeled with an Arrhenius-type temperature dependence.	$Q_{side0}(1 - SOC)e^{(-E_a/RT)}$	Eq. (S1.4), Supp. Mat.
Aging heat (Q_{aging})	Heat generated from aging-induced irreversible capacity loss and increased impedance.	$\alpha_{aging}\left(\frac{It}{Q_{max}}\right)^{0.4}$	Eq. (S1.5), Supp. Mat.
Post-discharge heat sink (Q_{stop})	Net endothermic effect from relaxation processes after current interruption (sum of five sub-mechanisms).	Sum of endothermic processes after current interruption	Eqs. (S1.6)–(S1.10), Supp. Mat.

Figure 8: Table of Different Heat Generation Components and Descriptions

The common source of heat, denoted in the table in **Figure 8**, that is shared with general electronic components is that of the "Joule effect" (Q_{Joule}), which is the heat loss based on the internal resistance of a component with its current, typically denoted as:

$$\dot{Q}_{joule} = I^2 R_{int},$$

Where I is the current flowing through the component, R_{int} is internal resistance, and t is the elapsed time [8]. In our battery box, these are the pre-charge/main contactors, main fuse, and BMS PCB boards, all of which have their own resistances and currents flowing through them. These simply transmit current to other components on the car, such as sensors, and their heat generation will be analyzed in future design reviews as these are outside of the scope of this report. We can use the $\dot{Q}_{gen}$ modeled to estimate the cell's temperature response under a lumped-capacitance assumption and eventually determine the necessary convection coefficient using:

$$mc_p \frac{dT}{dt} = \dot{Q_{gen}} - hA_s(T - T_s),$$

Where h is the convection coefficient, m is the mass of the battery, and c_p is the specific heat, T_s is the temperature of the surroundings, and T is the battery cell's temperature. With reactive cooling, the convection coefficient $h = f(I, V, T)$, where it is a function of BMS readings. Of course, in the most general case, where air flows across the stacks of batteries, we would estimate h with an appropriate Nusselt number with the general form:

$$h = \frac{N_{u_{D,N_L}} K_{air}}{D}$$

Where K_{air} is the thermal conductivity of air and D is the diameter of a single battery cell. As mentioned before, with reactive cooling, $N_{u_{D,N_L}} = f(I, V, T)$ because fan speeds would dictate Reynold's number of the fluid moving within battery cell modules, therefore changing the Nusselt number.

For lithium-ion batteries, thermal runaway (TR) is another critical issue that may arise, which is difficult to control when reached [9]. TR in lithium-ion batteries is a chain reaction of exothermic chemical reactions that lead battery cells' temperatures to spike dramatically, which can result in a battery fire or explosion. This is also the inspiration behind the need of accurate battery heat generation modeling to predict such an instant. In finding ways to mitigate TR, Wahab's research team validated a three-dimensional COMSOL model with forced-flow immersion cooling for an 18-cell lithium-ion battery pack [10]. They looked at variables such as cell spacing, coolant flow rate, and inlet temperature. In their modeling, increasing cell spacing was beneficial for optimal cooling. However, because they used immersion cooling, cell spacing will be a design variable since we do not use immersion cooling as a customer need states that there cannot be liquid inside the battery box, which is derived from racing regulations.

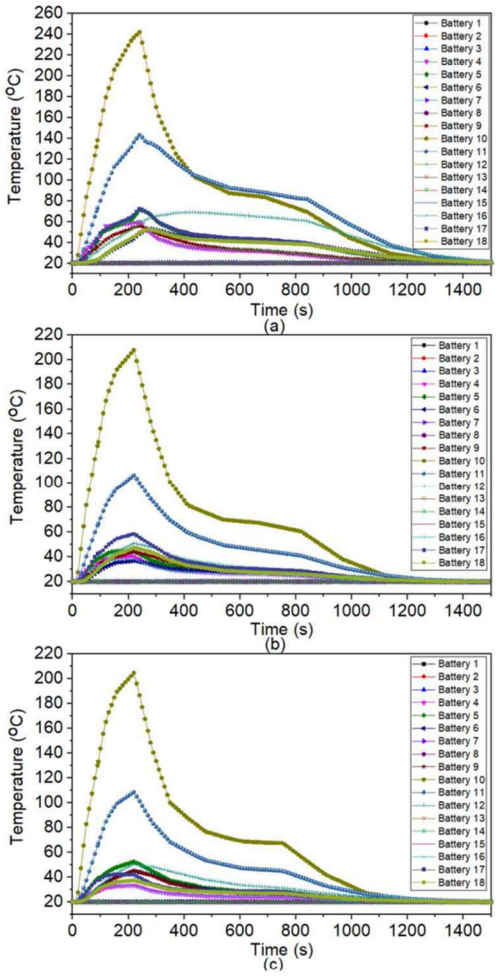


Figure 9: Plots of Temperature (°C) vs. Time (s) for increasing inter-cell spacing (a to c)

Figure 9 shows the difference in battery temperatures as inter-cell spacing increases from 1mm in plot (a) to 2 mm in plot (b). Therefore, although we want to keep our overall battery box lightweight, ensuring proper cell spacing will be beneficial for enhanced battery life and safety. They found that TR propagation was more likely and extreme as the defective cell was closer to the center and when cell spacing is decreased. As mentioned before, because this study used immersion cooling, its results do not directly prove an optimal spacing for our battery box since ours will be air-cooled (per customer needs/race regulations). Therefore, the purposes of their research was to inform about safety considerations. Combined with a predictive model discussed previously and forced cooling, which this research team argues for, an efficient BMS would be a great way to monitor cells. With these research teams having used COMSOL for thermal modeling, it would probably be beneficial to our team to also use this software because it specifically supports battery thermal modeling.

Problem

Our mission is to create a battery box to hold the batteries and control electronics for the Texas A&M University Solar Car while maintaining proper temperatures and ventilation in accordance with safety regulations. In doing so, our goals are to secure the battery system on a vehicle at speed, prevent damage to the battery and driver in a crash. Our stakeholders also guide our avenues for creative design, such as methods for modular placement of battery cells, air intake designs, battery cooling mechanisms, etc.

After meetings with our sponsors and considering goals, not methods, to keep our problem statement as neutral as possible, we have defined our solution-neutral problem statement (SNPS) as: making a battery box to cool, ventilate, and house batteries & other electronic components, while staying lightweight and inexpensive, for the Texas A&M University Solar Car in accordance with safety and racing regulations.

Having received the battery cell model we are to implement, temperature conditions have been set, giving us numerical limits for us to meet with our battery box. The problem is making sure the battery box components and other electronics inside do not overheat and therefore cause the car to malfunction or lose efficiency. As our sponsor pointed out, throughout the duration of the race, once the pack voltage drops below 90V, the motor controller draws more current to meet its power demand, for about constant power demand, therefore increasing heat generation. This becomes another condition to account for in modeling.

One of the implicit expectations from our sponsor is to make the battery box as compact as possible – this addresses both the issue of keeping it lightweight and allowing other sub-teams – battery, chassis, aeroshell, etc. – to have enough space for their own parts. A sponsor need is for our battery box to have an intake and outtake for air to flow through. Designing for this scenario would also help us meet the regulation of having negative pressure inside the battery box. Another implicit expectation is for the battery box with all its components to be lighter than previous year's car, which was 49 lbs. To make it more interactive with other sub-teams, an implicit expectation is to create the fluid flow and heat generation modeling with Ansys, which is the software commonly used by them. This report proposes using COMSOL for the battery heat generation. We seek to record the heat generation behavior on COMSOL and then set the battery pack heat generation localization and values in Ansys to show off the same properties. This would allow our team to model heat transfer and fluid dynamics in one program.

One of the inherent technical conflicts with the design task with my report would be analyzing the chemical reactions inside the batteries. A BMS is unable to do so as it only measures on the exterior of the cells, but my report proposes using equations that have been derived from known chemical processes/discharges to create an accurate thermal model.

Conclusion

Ideas for modeling battery box heat generation are discussed, where sources of heat were explored specific to lithium-ion batteries that must be accounted for. Discussed in this report and imperative to consider are specific instances and outside factors that may increase heat generation in battery cells and potential modes of failures. Battery management systems are used to measure operating variables such as voltage and temperature, but if accounting for the electrochemical effects and polarity changes within the battery, adjustments to battery usage and reactive cooling can be made preemptively to maintain efficiency as conditions degrade during the race. The usage of COMSOL software to model packs of battery cells in three-dimensions will be explored, where we can specify functions for the different sources of heat related to the battery cells we discussed – Siemens software remains an option. As far as concrete deliverables, we will specifically attempt to derive per-cell and total-pack heat generation rates, both as a function of time, and volumetric heat sources. We will also look at the results for selected conditions, such as acceleration, sustained battery performance/driving, and the sponsor's below 90-V condition. Assumptions we will include for each result are current/load, initial temperature, SoC, and operating duration.

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